

Git 3

1. Branching in Git

Branching is the process of creating a separate copy of the code to work on a new feature or bug fix without affecting the main branch.

Benefits:

- Keeps the main code safe.
- Allows multiple developers to work simultaneously.
- Makes feature development and bug fixing easier.

2. Merging in Git

Merging is the process of combining changes from one branch into another, usually from a feature branch into the main branch.

Benefits:

- Integrates completed work into the main project.
- Preserves all changes made in different branches.
- May require resolving merge conflicts if the same files were modified.

3. Creating a Branch in GitLab

A **branch** in GitLab is created to start working on a new feature or fix.

Steps:

1. Open your GitLab project.
2. Go to **Repository → Branches**.
3. Click **New Branch**.
4. Enter the branch name and select the source branch (e.g., main).
5. Click **Create Branch**.

4. Creating a Merge Request (MR) in GitLab

A **Merge Request (MR)** is used to request merging changes from one branch into another after code review.

Steps:

1. Push your branch to GitLab.
2. Go to **Merge Requests**.
3. Click **New Merge Request**.
4. Select the source and target branches.
5. Add a title and description.

6. Click **Create Merge Request**.
7. After review, click **Merge**.

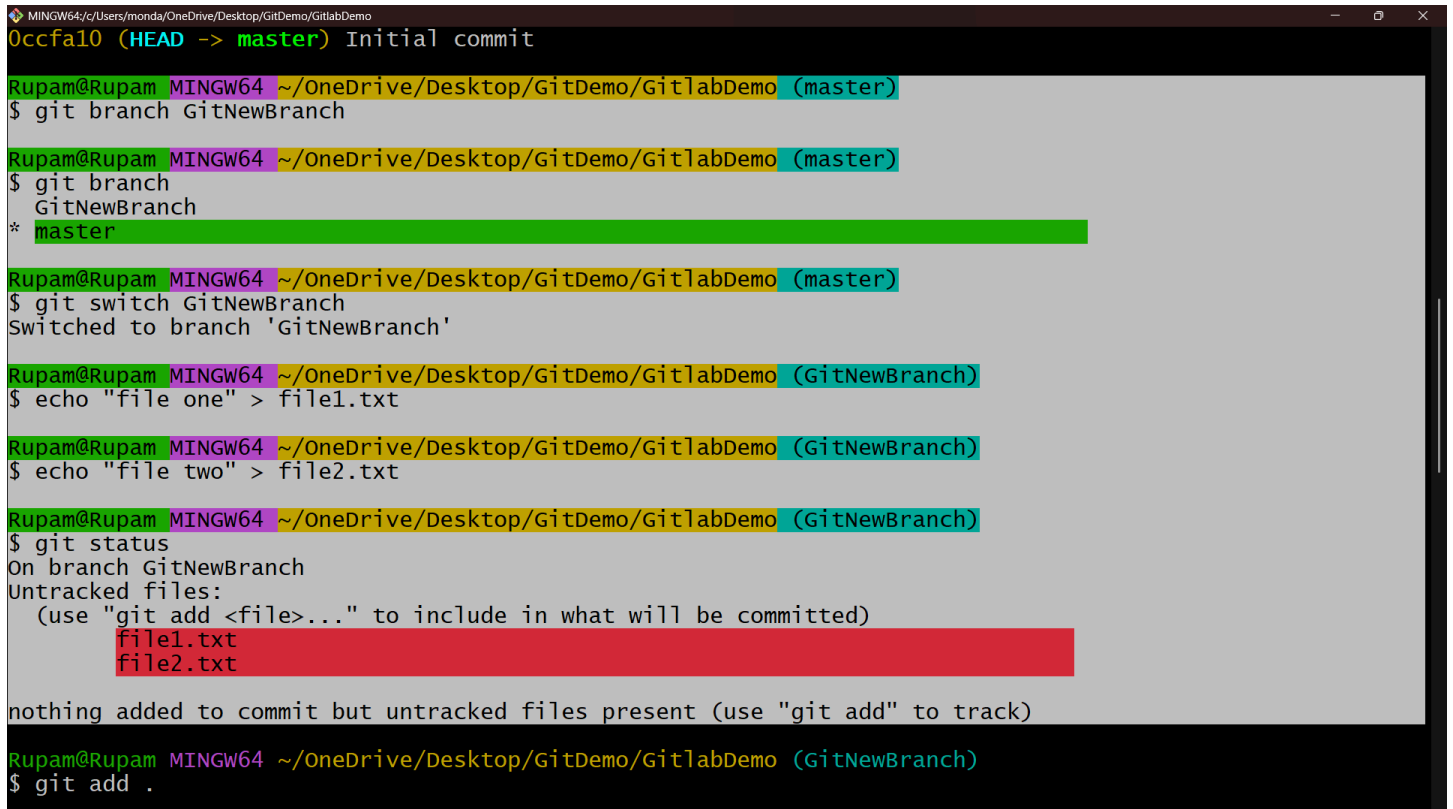
Hands ON

Branching

Creating GitNewBranch → using `git branch -c GitNewBranch`

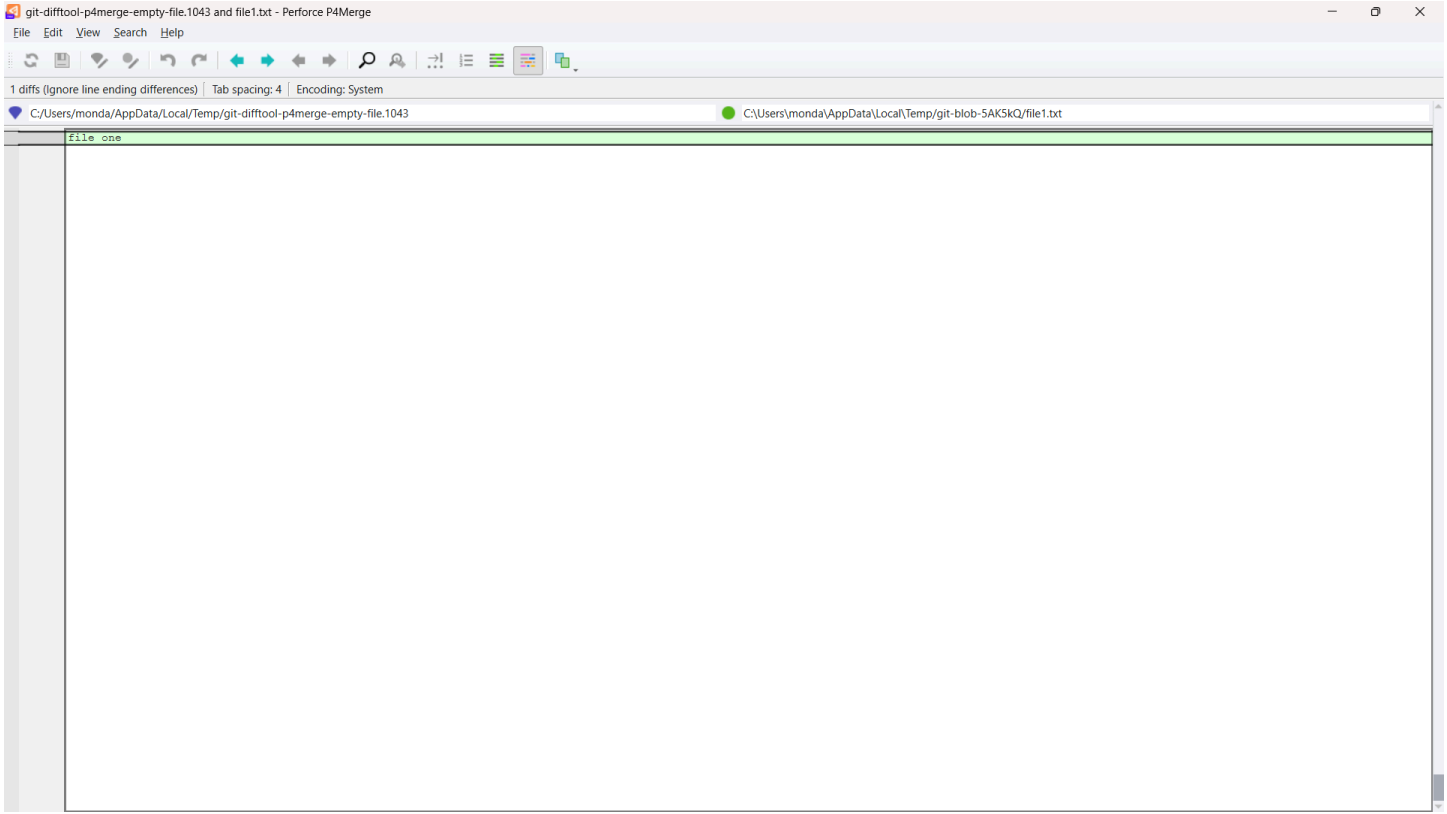
creating some contents in it

then using `git status` command , we see the status of tracked and untracked files .

A screenshot of a Windows terminal window with a dark background. The window title is 'MINGW64/c:/Users/monda/OneDrive/Desktop/GitDemo/GitlabDemo'. The terminal shows the following sequence of commands and outputs:
1. Initial commit: '0ccfa10 (HEAD -> master) Initial commit'.
2. Command: '\$ git branch GitNewBranch'.
3. Command: '\$ git branch'. Output: 'GitNewBranch' and '* master' (highlighted in green).
4. Command: '\$ git switch GitNewBranch'. Output: 'Switched to branch 'GitNewBranch''.
5. Command: '\$ echo "file one" > file1.txt'.
6. Command: '\$ echo "file two" > file2.txt'.
7. Command: '\$ git status'. Output: 'On branch GitNewBranch', 'Untracked files:', '(use "git add <file>..." to include in what will be committed)', 'file1.txt', 'file2.txt' (all three files listed are highlighted in red).
8. Summary: 'nothing added to commit but untracked files present (use "git add" to track)'.
9. Command: '\$ git add .'.

Merging

In p4tool we give merge request and merge main branch with GitNewBranch



After that in GitBash we run merge command

git merge GitNewBranch

then delete branch by using → git branch -d GitNewBranch

```
MINGW64~/c:/Users/monda/OneDrive/Desktop/GitDemo/GitlabDemo
Rupam@Rupam MINGW64 ~/OneDrive/Desktop/GitDemo/GitlabDemo (GitNewBranch)
$ git add .
warning: in the working copy of 'file1.txt', LF will be replaced by CRLF the ne
xt time git touches it
warning: in the working copy of 'file2.txt', LF will be replaced by CRLF the ne
xt time git touches it

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/GitDemo/GitlabDemo (GitNewBranch)
$ git commit -m "Second commit done"
[GitNewBranch 9046726] Second commit done
2 files changed, 2 insertions(+)
create mode 100644 file1.txt
create mode 100644 file2.txt

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/GitDemo/GitlabDemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/GitDemo/GitlabDemo (GitNewBranch)
$ git switch master
Switched to branch 'master'

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/GitDemo/GitlabDemo (master)
$ git diff master GitNewBranch
diff --git a/file1.txt b/file1.txt
new file mode 100644
index 0000000..ce2b3df
--- /dev/null
+++ b/file1.txt
@@ -0,0 +1 @@
+file one
diff --git a/file2.txt b/file2.txt
new file mode 100644
index 0000000..6c970db
--- /dev/null
+++ b/file2.txt
@@ -0,0 +1 @@
+file two

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/GitDemo/GitlabDemo (master)
$ |
```

```
MINGW64~/c:/Users/monda/OneDrive/Desktop/GitDemo/GitlabDemo
$ git status
On branch GitNewBranch
nothing to commit, working tree clean

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/GitDemo/GitlabDemo (GitNewBranch)
$ git switch master
Switched to branch 'master'

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/GitDemo/GitlabDemo (master)
$ git diff master GitNewBranch
diff --git a/file1.txt b/file1.txt
new file mode 100644
index 0000000..ce2b3df
--- /dev/null
+++ b/file1.txt
@@ -0,0 +1 @@
+file one
diff --git a/file2.txt b/file2.txt
new file mode 100644
index 0000000..6c970db
--- /dev/null
+++ b/file2.txt
@@ -0,0 +1 @@
+file two

Viewing (1/2): 'file1.txt'
Launch 'p4merge' [Y/n]? y
Updating 0ccfa10..9046726
Fast-forward
 file1.txt | 1 +
 file2.txt | 1 +
 2 files changed, 2 insertions(+)
 create mode 100644 file1.txt
 create mode 100644 file2.txt
* 9046726 (HEAD -> master, GitNewBranch) Second commit done
* 0ccfa10 Initial commit
Deleted branch GitNewBranch (was 9046726).
On branch master
nothing to commit, working tree clean
```